

Beginner Level

S. No	Name of the Programs
1	Create an application that takes the name from a text box and shows hello message along with the name entered in text box, when the user clicks the OK button.
2	Create a screen that has input boxes for User Name, Password, Address, Gender (radio buttons for male and female), Age (numeric), Date of Birth (Date Picket), State (Spinner) and a Submit button. On clicking the submit button, print all the data below the Submit Button. Use 1. Linear Layout 2. Relative Layout and 3. Grid Layout or Table Layout.
3	Develop an application that inserts some notifications into Notification area and whenever a notification is inserted, it should show a toast with details of the notification.
4	Develop an application that uses a menu with 3 options for dialing a number, opening a website and to send an SMS. On selecting an option, the appropriate action should be invoked using intents.
5	Create an application that uses a text file to store user names and passwords (tab separated fields and one record per line) When the user submits a login name and password through a screen, the details should be verified with the text file data and if they match, show a dialog saying that login is successful. Otherwise, show the dialog with Login Failed message
6	Create an activity to navigate to different activities through Intents- Implicit and Explicit
7	Create an activity to create option Menu.
8	Create an activity using Button Controls- CheckBox, RadioButtons, Radio Groups

1.) Create an application that takes the name from a text box and shows hello message along with the name entered in text box, when the user clicks the OK button.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp"
    android:gravity="center"
    android:background="#F5F5F5">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome App"
        android:textSize="32sp"
        android:textStyle="bold"
        android:textColor="#2196F3"
        android:layout_marginBottom="48dp" />

    <EditText
        android:id="@+id/etName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter your name"
        android:padding="16dp"
        android:background="@android:drawable/editbox_background"
        android:textSize="18sp"
        android:layout_marginBottom="24dp" />

    <Button
        android:id="@+id/btnOK"
        android:layout_width="200dp"
        android:layout_height="wrap_content"
        android:text="OK"
        android:textSize="18sp"
        android:background="#4CAF50"
        android:textColor="@android:color/white"
        android:padding="12dp" />

    <TextView
        android:id="@+id/tvMessage"
        android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
    android:textSize="20sp"
    android:textStyle="bold"
    android:textColor="#FF5722"
    android:layout_marginTop="32dp" />
```

```
</LinearLayout>
```

mainActivity.java

```
package com.example.helloapp;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private EditText editTextName;
    private Button btnOK;
    private TextView textViewMessage;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        editTextName = findViewById(R.id.editTextName);
        btnOK = findViewById(R.id.btnOK);
        textViewMessage = findViewById(R.id.textViewMessage);

        btnOK.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String name = editTextName.getText().toString();
                if (!name.isEmpty()) {
                    textViewMessage.setText("Hello " + name + "!");
                } else {
                    textViewMessage.setText("Please enter your name");
                }
            }
        });
    }
}
```

2.) Create a screen that has input boxes for User Name, Password, Address, Gender (radio buttons for male and female), Age (numeric), Date of Birth (Date Picket), State (Spinner) and a Submit button. On clicking the submit button, print all the data below the Submit Button. Use

1. Linear Layout

2. Relative Layout and

3. Grid Layout or Table Layout.

activity_linear_layout.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">

        <EditText
            android:id="@+id/etUserName"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="User Name"
            android:layout_marginBottom="12dp"/>

        <EditText
            android:id="@+id/etPassword"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Password"
            android:inputType="textPassword"
            android:layout_marginBottom="12dp"/>

        <EditText
            android:id="@+id/etAddress"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Address"
            android:layout_marginBottom="12dp"/>

        <TextView
```

```
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Gender:"
    android:layout_marginBottom="8dp"/>
```

```
<RadioGroup
    android:id="@+id/radioGroupGender"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_marginBottom="12dp">
```

```
<RadioButton
    android:id="@+id/rbMale"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Male"/>
```

```
<RadioButton
    android:id="@+id/rbFemale"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Female"/>
```

```
</RadioGroup>
```

```
<EditText
    android:id="@+id/etAge"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Age"
    android:inputType="number"
    android:layout_marginBottom="12dp"/>
```

```
<EditText
    android:id="@+id/etDOB"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Date of Birth (DD/MM/YYYY)"
    android:layout_marginBottom="12dp"/>
```

```
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="State:"
    android:layout_marginBottom="8dp"/>
```

```
<Spinner
```

```

        android:id="@+id/spinnerState"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="16dp"/>

<Button
    android:id="@+id/btnSubmit"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Submit"/>

<TextView
    android:id="@+id/tvOutput"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="16dp"
    android:textSize="14sp"/>

</LinearLayout>
</ScrollView>

```

activity_relative_layout.xml

```

<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <RelativeLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:padding="16dp">

        <EditText
            android:id="@+id/etUserName"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="User Name"
            android:layout_alignParentTop="true"/>

        <EditText
            android:id="@+id/etPassword"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Password"
            android:inputType="textPassword"
            android:layout_below="@id/etUserName"
            android:layout_marginTop="12dp"/>
    </RelativeLayout>
</ScrollView>

```

```
<EditText
    android:id="@+id/etAddress"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Address"
    android:layout_below="@id/etPassword"
    android:layout_marginTop="12dp"/>
```

```
<TextView
    android:id="@+id/tvGender"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Gender:"
    android:layout_below="@id/etAddress"
    android:layout_marginTop="12dp"/>
```

```
<RadioGroup
    android:id="@+id/radioGroupGender"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_below="@id/tvGender"
    android:layout_marginTop="8dp">
```

```
<RadioButton
    android:id="@+id/rbMale"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Male"/>
```

```
<RadioButton
    android:id="@+id/rbFemale"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Female"/>
```

```
</RadioGroup>
```

```
<EditText
    android:id="@+id/etAge"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Age"
    android:inputType="number"
    android:layout_below="@id/radioGroupGender"
    android:layout_marginTop="12dp"/>
```

```

<EditText
    android:id="@+id/etDOB"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Date of Birth"
    android:layout_below="@id/etAge"
    android:layout_marginTop="12dp"/>

<TextView
    android:id="@+id/tvState"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="State:"
    android:layout_below="@id/etDOB"
    android:layout_marginTop="12dp"/>

<Spinner
    android:id="@+id/spinnerState"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@id/tvState"
    android:layout_marginTop="8dp"/>

<Button
    android:id="@+id/btnSubmit"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Submit"
    android:layout_below="@id/spinnerState"
    android:layout_marginTop="16dp"/>

<TextView
    android:id="@+id/tvOutput"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@id/btnSubmit"
    android:layout_marginTop="16dp"
    android:textSize="14sp"/>

```

```

</RelativeLayout>
</ScrollView>

```

activity_table_layout.xml

```

<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

```



```
<TableLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:padding="16dp"
    android:stretchColumns="1">

    <TableRow>
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="User Name:"/>
        <EditText
            android:id="@+id/etUserName"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Enter User Name"/>
    </TableRow>

    <TableRow>
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Password:"/>
        <EditText
            android:id="@+id/etPassword"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Enter Password"
            android:inputType="textPassword"/>
    </TableRow>

    <TableRow>
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Address:"/>
        <EditText
            android:id="@+id/etAddress"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Enter Address"/>
    </TableRow>

    <TableRow>
        <TextView
            android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:text="Gender:"/>
    <RadioGroup
        android:id="@+id/radioGroupGender"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">
        <RadioButton
            android:id="@+id/rbMale"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Male"/>
        <RadioButton
            android:id="@+id/rbFemale"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Female"/>
    </RadioGroup>
</TableRow>
```

```
<TableRow>
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Age:"/>
    <EditText
        android:id="@+id/etAge"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Age"
        android:inputType="number"/>
</TableRow>
```

```
<TableRow>
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Date of Birth:"/>
    <EditText
        android:id="@+id/etDOB"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="DD/MM/YYYY"/>
</TableRow>
```

```
<TableRow>
    <TextView
```

```

        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="State:"/>
    <Spinner
        android:id="@+id/spinnerState"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
</TableRow>

<TableRow>
    <Button
        android:id="@+id/btnSubmit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Submit"
        android:layout_span="2"/>
</TableRow>

</TableLayout>

<TextView
    android:id="@+id/tvOutput"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="16dp"
    android:padding="16dp"
    android:textSize="14sp"/>
</ScrollView>

Java
package com.example.registrationapp;

import android.os.Bundle;
import android.view.View;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class RegistrationActivity extends AppCompatActivity {

    private EditText etUserName, etPassword, etAddress, etAge, etDOB;
    private RadioGroup radioGroupGender;
    private Spinner spinnerState;
    private Button btnSubmit;
    private TextView tvOutput;
    private String selectedGender = "";

    @Override

```

```

protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_linear_layout); // Change layout name as needed

    initializeViews();
    setupSpinner();
    setupSubmitButton();
}

private void initializeViews() {
    etUserName = findViewById(R.id.etUserName);
    etPassword = findViewById(R.id.etPassword);
    etAddress = findViewById(R.id.etAddress);
    etAge = findViewById(R.id.etAge);
    etDOB = findViewById(R.id.etDOB);
    radioGroupGender = findViewById(R.id.radioGroupGender);
    spinnerState = findViewById(R.id.spinnerState);
    btnSubmit = findViewById(R.id.btnSubmit);
    tvOutput = findViewById(R.id.tvOutput);

    radioGroupGender.setOnCheckedChangeListener((group, checkedId) -> {
        if (checkedId == R.id.rbMale) {
            selectedGender = "Male";
        } else if (checkedId == R.id.rbFemale) {
            selectedGender = "Female";
        }
    });
}

private void setupSpinner() {
    String[] states = {"Select State", "California", "Texas", "New York", "Florida", "Illinois",
"Ohio"};
    ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
android.R.layout.simple_spinner_item, states);
    adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
    spinnerState.setAdapter(adapter);
}

private void setupSubmitButton() {
    btnSubmit.setOnClickListener(v -> {
        String userName = etUserName.getText().toString();
        String password = etPassword.getText().toString();
        String address = etAddress.getText().toString();
        String age = etAge.getText().toString();
        String dob = etDOB.getText().toString();
        String state = spinnerState.getSelectedItem().toString();
    });
}

```

```

        String output = "User Name: " + userName + "\n" +
            "Password: " + password + "\n" +
            "Address: " + address + "\n" +
            "Gender: " + selectedGender + "\n" +
            "Age: " + age + "\n" +
            "Date of Birth: " + dob + "\n" +
            "State: " + state;

        tvOutput.setText(output);
    });
}
}

```

3.) Develop an application that inserts some notifications into Notification area and whenever a notification is inserted, it should show a toast with details of the notification.

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <EditText
        android:id="@+id/etNotificationTitle"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Notification Title"
        android:layout_marginBottom="12dp"/>

    <EditText
        android:id="@+id/etNotificationMessage"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Notification Message"
        android:layout_marginBottom="24dp"/>

    <Button
        android:id="@+id/btnSendNotification"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send Notification"/>

```

</LinearLayout>

MainActivity.java

```
package com.example.notificationapp;

import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.content.Context;
import android.os.Build;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.NotificationCompat;

public class MainActivity extends AppCompatActivity {

    private static final String CHANNEL_ID = "NOTIFICATION_CHANNEL";
    private static final int NOTIFICATION_ID = 1;

    private EditText etTitle, etMessage;
    private Button btnSendNotification;
    private NotificationManager notificationManager;
    private int notificationCounter = 1;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etTitle = findViewById(R.id.etNotificationTitle);
        etMessage = findViewById(R.id.etNotificationMessage);
        btnSendNotification = findViewById(R.id.btnSendNotification);

        notificationManager = (NotificationManager)
getSystemService(Context.NOTIFICATION_SERVICE);

        createNotificationChannel();

        btnSendNotification.setOnClickListener(v -> {
            String title = etTitle.getText().toString();
            String message = etMessage.getText().toString();

            if (!title.isEmpty() && !message.isEmpty()) {
```

```

        sendNotification(title, message);
        Toast.makeText(MainActivity.this,
            "Notification Sent: " + title, Toast.LENGTH_LONG).show();
    } else {
        Toast.makeText(MainActivity.this,
            "Please enter both title and message", Toast.LENGTH_SHORT).show();
    }
    });
}

private void createNotificationChannel() {
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
        CharSequence name = "My Notification Channel";
        String description = "Channel for my notifications";
        int importance = NotificationManager.IMPORTANCE_DEFAULT;
        NotificationChannel channel = new NotificationChannel(CHANNEL_ID, name,
importance);
        channel.setDescription(description);
        notificationManager.createNotificationChannel(channel);
    }
}

private void sendNotification(String title, String message) {
    NotificationCompat.Builder builder = new NotificationCompat.Builder(this,
CHANNEL_ID)
        .setSmallIcon(android.R.drawable.ic_dialog_info)
        .setContentTitle(title)
        .setContentText(message)
        .setPriority(NotificationCompat.PRIORITY_DEFAULT)
        .setAutoCancel(true);

    notificationManager.notify(notificationCounter++, builder.build());
}
}

```

4.) Develop an application that uses a menu with 3 options for dialing a number, opening a website and to send an SMS. On selecting an option, the appropriate action should be invoked using intents.

menu_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto">

```

```

<item
    android:id="@+id/action_dial"
    android:title="Dial Number"
    android:icon="@android:drawable/ic_menu_call"
    app:showAsAction="ifRoom"/>

<item
    android:id="@+id/action_website"
    android:title="Open Website"
    android:icon="@android:drawable/ic_menu_view"
    app:showAsAction="ifRoom"/>

<item
    android:id="@+id/action_sms"
    android:title="Send SMS"
    android:icon="@android:drawable/ic_menu_send"
    app:showAsAction="ifRoom"/>

</menu>

```

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Menu Demo App"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginBottom="24dp"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Click the menu icon to see options"
        android:textSize="16sp"/>

</LinearLayout>

```


MainActivity.java

```
package com.example.menuapp;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.menu_main, menu);
        return true;
    }

    @Override
    public boolean onOptionsItemSelected(MenuItem item) {
        int id = item.getItemId();

        if (id == R.id.action_dial) {
            dialNumber();
            return true;
        } else if (id == R.id.action_website) {
            openWebsite();
            return true;
        } else if (id == R.id.action_sms) {
            sendSMS();
            return true;
        }

        return super.onOptionsItemSelected(item);
    }

    private void dialNumber() {
        Intent intent = new Intent(Intent.ACTION_DIAL);
        intent.setData(Uri.parse("tel:1234567890"));
        startActivity(intent);
    }
}
```

```

    }

    private void openWebsite() {
        Intent intent = new Intent(Intent.ACTION_VIEW);
        intent.setData(Uri.parse("https://www.google.com"));
        startActivity(intent);
    }

    private void sendSMS() {
        Intent intent = new Intent(Intent.ACTION_VIEW);
        intent.setData(Uri.parse("sms:1234567890"));
        intent.putExtra("sms_body", "Hello from my app!");
        startActivity(intent);
    }
}

```

5 & 6) Create an application that uses a text file to store user names and passwords (tab separated fields and one record per line) When the user submits a login name and password through a screen, the details should be verified with the text file data and if they match, show a dialog saying that login is successful. Otherwise, show the dialog with Login Failed message

Create an activity to navigate to different activities through Intents- Implicit and Explicit

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Login"
        android:textSize="28sp"
        android:textStyle="bold"
        android:layout_marginBottom="32dp"/>

    <EditText

```

```

        android:id="@+id/etUsername"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Username"
        android:layout_marginBottom="16dp"/>

<EditText
    android:id="@+id/etPassword"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Password"
    android:inputType="textPassword"
    android:layout_marginBottom="24dp"/>

<Button
    android:id="@+id/btnLogin"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Login"/>

</LinearLayout>

```

MainActivity.java

```

package com.example.loginapp;

import android.content.Context;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import java.io.BufferedReader;
import java.io.FileOutputStream;
import java.io.InputStream;
import java.io.InputStreamReader;
import java.io.OutputStreamWriter;

public class MainActivity extends AppCompatActivity {

    private EditText etUsername, etPassword;
    private Button btnLogin;
    private static final String FILENAME = "users.txt";

    @Override
    protected void onCreate(Bundle savedInstanceState) {

```

```

super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);

etUsername = findViewById(R.id.etUsername);
etPassword = findViewById(R.id.etPassword);
btnLogin = findViewById(R.id.btnLogin);

createSampleUserFile();

btnLogin.setOnClickListener(v -> {
    String username = etUsername.getText().toString().trim();
    String password = etPassword.getText().toString().trim();

    if (validateUser(username, password)) {
        showDialog("Login Successful", "Welcome " + username + "!");
    } else {
        showDialog("Login Failed", "Invalid username or password!");
    }
});
}

private void createSampleUserFile() {
    try {
        FileOutputStream fos = openFileOutput(FILENAME, Context.MODE_PRIVATE);
        OutputStreamWriter osw = new OutputStreamWriter(fos);
        osw.write("admin\tadmin123\n");
        osw.write("user1\tpassword1\n");
        osw.write("john\tjohn123\n");
        osw.write("jane\tjane123\n");
        osw.close();
        fos.close();
    } catch (Exception e) {
        e.printStackTrace();
    }
}

private boolean validateUser(String username, String password) {
    try {
        InputStream is = openFileInput(FILENAME);
        BufferedReader reader = new BufferedReader(new InputStreamReader(is));
        String line;

        while ((line = reader.readLine()) != null) {
            String[] parts = line.split("\t");
            if (parts.length == 2) {
                if (parts[0].equals(username) && parts[1].equals(password)) {
                    reader.close();
                    return true;
                }
            }
        }
        return false;
    } catch (Exception e) {
        e.printStackTrace();
        return false;
    }
}

```

```

        return true;
    }
}
}
reader.close();
} catch (Exception e) {
    e.printStackTrace();
}
return false;
}

private void showDialog(String title, String message) {
    new AlertDialog.Builder(this)
        .setTitle(title)
        .setMessage(message)
        .setPositiveButton("OK", (dialog, which) -> dialog.dismiss())
        .show();
}
}

```

7.) Create an activity to create option Menu.

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Intent Demo"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginBottom="32dp"/>

    <Button
        android:id="@+id/btnExplicitIntent"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Explicit Intent"
        android:layout_marginBottom="16dp"/>

```

```
<Button
    android:id="@+id/btnImplicitDial"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Implicit Intent - Dial"
    android:layout_marginBottom="16dp"/>
```

```
<Button
    android:id="@+id/btnImplicitWeb"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Implicit Intent - Website"/>
```

```
</LinearLayout>
```

activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">
```

```
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Second Activity"
    android:textSize="24sp"
    android:textStyle="bold"
    android:layout_marginBottom="16dp"/>
```

```
<TextView
    android:id="@+id/tvMessage"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="You reached the second activity!"
    android:textSize="16sp"/>
```

```
</LinearLayout>
```

SecondActivity.java

```
package com.example.intentapp;
```

```
import android.os.Bundle;
```

```
import android.widget.TextView;
```

```

import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);

        TextView tvMessage = findViewById(R.id.tvMessage);

        String receivedData = getIntent().getStringExtra("DATA");
        if (receivedData != null) {
            tvMessage.setText(receivedData);
        }
    }
}

```

MainActivity.java

```

package com.example.intentapp;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private Button btnExplicit, btnImplicitDial, btnImplicitWeb;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnExplicit = findViewById(R.id.btnExplicitIntent);
        btnImplicitDial = findViewById(R.id.btnImplicitDial);
        btnImplicitWeb = findViewById(R.id.btnImplicitWeb);

        btnExplicit.setOnClickListener(v -> {
            Intent intent = new Intent(MainActivity.this, SecondActivity.class);
            intent.putExtra("DATA", "Hello from Main Activity!");
            startActivity(intent);
        });
    }
}

```

```

        btnImplicitDial.setOnClickListener(v -> {
            Intent intent = new Intent(Intent.ACTION_DIAL);
            intent.setData(Uri.parse("tel:1234567890"));
            startActivity(intent);
        });

        btnImplicitWeb.setOnClickListener(v -> {
            Intent intent = new Intent(Intent.ACTION_VIEW);
            intent.setData(Uri.parse("https://www.google.com"));
            startActivity(intent);
        });
    }
}

```

8.) Create an activity using Button Controls- CheckBox, RadioButtons, Radio Groups.

menu_options.xml

```

<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto">

    <item
        android:id="@+id/action_home"
        android:title="Home"
        android:icon="@android:drawable/ic_menu_home"
        app:showAsAction="ifRoom"/>

    <item
        android:id="@+id/action_search"
        android:title="Search"
        android:icon="@android:drawable/ic_menu_search"
        app:showAsAction="ifRoom"/>

    <item
        android:id="@+id/action_settings"
        android:title="Settings"
        android:icon="@android:drawable/ic_menu_preferences"
        app:showAsAction="never"/>

    <item
        android:id="@+id/action_help"
        android:title="Help"
        android:icon="@android:drawable/ic_menu_help"
        app:showAsAction="never"/>

```



```
<item
    android:id="@+id/action_about"
    android:title="About"
    app:showAsAction="never"/>

</menu>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Option Menu Demo"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginBottom="16dp"/>

    <TextView
        android:id="@+id/tvMessage"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Click the menu icon (three dots) to see options"
        android:textSize="16sp"
        android:gravity="center"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.optionmenuapp;

import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private TextView tvMessage;
```

```

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    tvMessage = findViewById(R.id.tvMessage);
}

@Override
public boolean onCreateOptionsMenu(Menu menu) {
    getMenuInflater().inflate(R.menu.menu_options, menu);
    return true;
}

@Override
public boolean onOptionsItemSelected(MenuItem item) {
    int id = item.getItemId();

    if (id == R.id.action_home) {
        tvMessage.setText("Home selected");
        Toast.makeText(this, "Home clicked", Toast.LENGTH_SHORT).show();
        return true;
    } else if (id == R.id.action_search) {
        tvMessage.setText("Search selected");
        Toast.makeText(this, "Search clicked", Toast.LENGTH_SHORT).show();
        return true;
    } else if (id == R.id.action_settings) {
        tvMessage.setText("Settings selected");
        Toast.makeText(this, "Settings clicked", Toast.LENGTH_SHORT).show();
        return true;
    } else if (id == R.id.action_help) {
        tvMessage.setText("Help selected");
        Toast.makeText(this, "Help clicked", Toast.LENGTH_SHORT).show();
        return true;
    } else if (id == R.id.action_about) {
        tvMessage.setText("About selected");
        Toast.makeText(this, "About clicked - Version 1.0", Toast.LENGTH_LONG).show();
        return true;
    }

    return super.onOptionsItemSelected(item);
}
}

```